Study Notes Search

A small AI-assisted development prototype for searching study notes using classical ML text retrieval. This is not a generative chatbot or semantic embedding search.

Features

- Paste notes and enter a question.
- Split notes into passages, represent them with TF-IDF, and rank with cosine similarity.
- Display source passages and similarity scores.
- Handle empty input and no matching vocabulary.
- Download retrieved passages as text.

Run locally

Install Python 3.11 or newer, then run:

```
pip install -r requirements.txt
streamlit run app.py
```

Deploy

Put app.py, requirements.txt, and README.md in the root of a public GitHub repository. On Streamlit Community Cloud, create an app, choose that repository, and select app.py as the entrypoint.

Manual checks

1. With demo notes, ask: How does TCP provide reliable delivery? The TCP passage should rank first.
2. Ask: What is a primary key? The relational database passage should rank first.
3. Ask: banana spaceship. A no-matching-terms message should appear.
4. Empty either input: the app should request both fields.
5. Download results and compare them with the displayed passages.

Design trade-off

Retrieving source passages is easier to inspect than generating answers, and requires no model API key. The trade-off is limited language understanding: lexical matching can miss synonyms, and similarity is not proof that a passage answers a question. This design rationale comes with the starter project; explain your own changes and decisions honestly.

Attribution and status

Initial code generated with an AI assistant. Add your own implementation changes, test results, and dates here. Do not claim independent authorship or production readiness. Do not claim tests passed until you have run them.